SC3102: Signals, Systems, and Transforms

Tutorial 02: DFT and Discrete Fourier Transforms

1. Fix $N = 4$.

(a) Compute the primitive root of unity and write out the analysis and synthesis matrices.

(b) What do you observe about the 1st and 3rd rows of the two matrices (0 based indexing)?

(c) If the input vector, x, is real, what can you conclude about its DFT representation X? Specifically, comment on the DFT coefficients $X(1)$ and $X(3)$.

(d) Suppose we measured a real vector v and its DFT representation V satisfying:

$$v = \begin{bmatrix} v[0] \\ v[1] \\ 0 \\ v[3] \end{bmatrix} \qquad V = \begin{bmatrix} 4 \\ 3+3j \\ A \\ B \end{bmatrix}.$$

Use the conclusions of the previous parts to determine A, B and v.

2. For this question, consider the two signals:

$$a[n] = \cos\left(2\pi\frac{n}{8}\right) \text{ and } b[n] = \sin\left(2\pi\frac{n}{6} + \frac{\pi}{3}\right)$$

(a) Compute the fundamental periods of a and b respectively?

(b) Without using the formula, compute the DTFS of a and b. Use the fundamental periods computed in the previous question.

(c) Suppose, the DTFS for both signals was computed using the first $N = 12$ indices $\{0, \ldots, 11\}$. Does the synthesis equation applied to this 12 point DTFS recover the original signals for all n? Justify your answer.

3. Consider the aperiodic signals:

$$x[n] = a^{-n}u[n] \text{ and } y[n] = (n+1)a^{-n}u[n].$$

For these signals, answer the following questions:

(a) For what values of a are the DTFT of x and y defined?

(b) When the DTFT is defined, what are the DTFTs of x and y? How are they related?

(c) In practical settings, we do not have access to all the entries of an input signal. Suppose we approximated the DTFT with a finite number of samples:

$$X_N(e^{j\Omega}) = \sum_{n=0}^{N} x[n]e^{-j\Omega n}.$$

How many samples do we need for $\left|X_N(e^{j\Omega}) - X(e^{j\Omega})\right| \leq 0.01$ for all Ω when $a = 2$? What happens when a approaches 1 from above?

4. For this question, consider the two signals with the first entries below:

$$x = [2, -1, 1, 4] \text{ and } y = [-1, 0, 2].$$

The first entries correspond to $n = 0$ and the signal value is 0 outside the listed indices.

(a) First, compute the linear convolution of x and y.

(b) Let X and Y denote the DFT of x and y for $N = 4$. Without explicitly computing the DFTs, what is the inverse DFT of the product of X and Y?

(c) How can you obtain the result of the circular convolution in the previous part from the linear convolution? What is the relationship between them?

(d) How many zeros should you pad to the ends of the two signals for the circular convolution to recover the linear convolution? Justify why in terms of the relationship derived in the previous part.